



Multi-step Traffic Flow Prediction Based on Reinforced Attention Graph Convolutional Network

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Abstract. Accurate multi-step traffic flow prediction plays a crucial role in intelligent transportation systems, enabling efficient traffic management and congestion mitigation. However, existing sequence-to-sequence models often fail to capture dynamic spatiotemporal dependencies, particularly struggling with evolving traffic network topologies and long-range temporal correlations. To tackle these challenges, we propose a sequence-to-sequence model based on Reinforced Attention Graph Convolutional Networks to comprehensively utilize the spatiotemporal dependencies in traffic data. Specifically, we integrate Graph Convolutional Networks with Gated Recurrent Units to enhance spatiotemporal feature extraction capabilities. On the decoder side, we propose an exponential-decay temporal attention mechanism that effectively captures features from distant time steps. Additionally, we develop an input-feedback-based training strategy to boost the model's capacity to retain historical attention information, effectively mitigating error accumulation and enhancing prediction accuracy. Extensive experiments on two real-world traffic datasets, SZ-taxi and Los-Loop, confirm the superiority of the proposed model over existing baseline methods in multi-step traffic flow prediction.

Keywords: Traffic Flow Prediction · Graph Convolutional Network · Spatio-temporal · Attention Mechanism

1 Introduction

With rapid urbanization and growing vehicle numbers, Intelligent Transportation Systems (ITS) play a critical role in improving traffic efficiency and reducing congestion. Among them, traffic flow prediction serves as a cornerstone for key applications such as adaptive signal control, route planning, and resource allocation [1]. Different from other time series prediction problems, traffic flow prediction involves complex spatiotemporal dependencies stemming from road network topology and dynamic traffic patterns. Traffic data exhibit strong temporal periodicity and spatial interactions across distant segments, necessitating joint modeling of both spatial and temporal features.

Deep learning methods, especially Sequence-to-Sequence (Seq2Seq) models [2], have shown promise in modeling temporal dependencies through attention mechanisms. However, many such models overlook dynamic spatial correlations and adopt loosely coupled Convolutional Recurrent networks [3, 4] that fail to capture tightly integrated spatiotemporal patterns. To address this, researchers have proposed unified architectures, such as ST-ResNet [5] and ConvLSTM [6], which integrate temporal and spatial modeling. These methods have significantly improved the accuracy of traffic evolution pattern modeling. However, several limitations remain: 1) Most models rely on static graph structures for spatial modeling, limiting adaptability to dynamic traffic topologies. 2) Existing attention mechanisms fail to adequately consider temporal information in weight calculation. 3) Contextual dependencies across time steps are frequently diminished in long-sequence predictions.

To address the above issues, we propose a multi-step traffic flow prediction model based on a Reinforced Attention Graph Convolutional Sequence-to-Sequence framework (RAGC-Seq2Seq). The model's encoder unit employs a hybrid architecture combining GCNs and GRUs to jointly model spatial dependencies and temporal dynamics. Meanwhile, an exponential-decay temporal attention mechanism is integrated into the traditional Luong attention mechanism, significantly improving long-term temporal dependency modeling. Additionally, an input-feedback strategy is incorporated during the decoding phase to reduce error propagation.

2 Related Work

2.1 Temporal Modeling for Traffic Flow Prediction

Time series-based traffic flow prediction primarily focuses on modeling nonlinear sequences and capturing long-term temporal dependencies. Early studies employed statistical approaches, represented by the AutoRegressive Integrated Moving Average (ARIMA) model [7]. These approaches typically assume that traffic flow data are stationary or follow linear patterns. Subsequently, machine learning approaches like support vector regression (SVR) [8] were introduced to model nonlinear patterns by mapping data to high-dimensional spaces. While these methods improve prediction accuracy to some extent, they rely on manual feature engineering and struggle with scalability in large spatiotemporal datasets.

The emergence of deep learning has revolutionized traffic prediction due to their automatic feature extraction and nonlinear mapping capabilities. Representative models include LSTMs [9] and GRUs [10], which excel at capturing temporal dependencies.

However, these models suffer from vanishing gradients and limited long-term memory due to their sequential nature. To address this, WaveNet [11] enhances nonlinear modeling with gated convolutions. Informer [12] introduces probabilistic sparse self-attention, achieving optimal performance in traffic forecasting. Nevertheless, most of these approaches largely neglect the spatial dependencies embedded in traffic networks.

2.2 Spatiotemporal Modeling for Traffic Flow Prediction

Spatiotemporal traffic prediction requires simultaneous modeling of temporal dynamics and spatial dependencies in road networks. Early studies often employed a divide-and-conquer approach to model spatiotemporal features. ConvLSTM [6] employed convolutional LSTM units to extract local spatiotemporal features, but its grid-based convolution kernels struggle to represent irregular road topologies. To address this, ST-ResNet [5] stacks 3D convolutional layers in a residual network to capture multiscale spatiotemporal dependencies. However, the use of fixed-scale kernels limits its adaptability to dynamic topological variations.

To better handle dynamic traffic interactions, adaptive graph learning techniques have been proposed. For example, Liu et al. [13] developed a dynamic spatiotemporal graph convolutional network with interactive learning, and tensor decomposition methods [14] extract latent patterns via low-rank approximations. However, these approaches are typically designed for fixed-horizon prediction and struggle with long-sequence generation. DCRNN [15] adopts an encoder-decoder structure to capture spatiotemporal dependencies, but its static context transmission limits long-term modeling. To overcome this, attention mechanisms have been widely integrated into Seq2Seq architectures. Kong et al. [16] proposed a variational Bayesian encoder-decoder framework that incorporates probabilistic attention to reduce the impact of observational noise via variational inference.

3 Preliminaries

Definition 1 (Traffic Road Network). We represent the topological structure of the traffic road network using an unweighted and undirected graph $G = (V, E, A)$, where $V = v_1, v_2, \dots, v_N$ denotes the set of nodes, each representing a road segment, and N is the total number of nodes. E encodes topological connections between nodes. $A \in \mathbb{R}^{N \times N}$ represents the adjacency matrix of G , characterizing spatial correlations between nodes.

Definition 2 (Traffic Flow Features). At time step t , the traffic feature of node v_t is denoted as $x_t^i \in \mathbb{R}^C$, where C indicates the dimensionality of traffic features, such as speed and volume. The network-wide traffic state at t is formulated as a matrix $X_t = (x_t^1, x_t^2, \dots, x_t^{N-1}, x_t^N) \in \mathbb{R}^{N \times C}$. Considering the past T time steps, the historical traffic feature sequence is denoted by $X_{(t-T+1):t} \in \mathbb{R}^{N \times T \times C}$. Similarly, the predicted sequence for M future steps is $\hat{X}_{(t+1):(t+M)} \in \mathbb{R}^{N \times M \times C}$.

Definition 3 (Multi-step Traffic Flow Prediction). Given the historical traffic features over the past T time steps $X = (X_{t-T+1}, X_{t-T+2}, \dots, X_t) \in \mathbb{R}^{N \times T}$, the objective is to predict the traffic conditions over the subsequent Q time steps $\hat{X}_{t+1}, \hat{X}_{t+2}, \dots, \hat{X}_{t+Q}$. The joint distribution can be expressed as:

$$P(\hat{X}_{t+1}, \hat{X}_{t+2}, \dots, \hat{X}_{t+Q} | X_{t-T+1}, \dots, X_t) = \prod_{i=1}^Q P(\hat{X}_{t+i} | C_t, \hat{X}_{t+1}, \hat{X}_{t+2}, \dots, \hat{X}_{t+i-1}) \quad (1)$$

where C_i denotes the context representation of the input sequence X . $P(\cdot)$ represents the conditional probability function, and Q is the number of prediction steps.

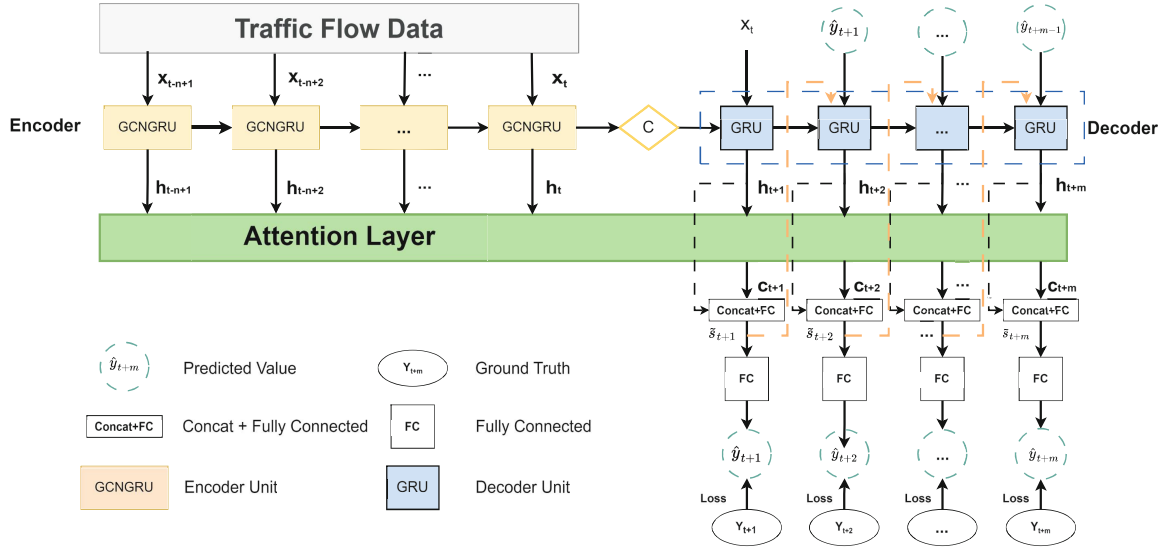


Fig. 1. Overall framework of the RAGC-Seq2Seq model

4 Methodology

Our proposed RAGC-Seq2Seq framework uniquely integrates spatiotemporal modeling within a sequence-to-sequence paradigm for traffic prediction. The overall architecture is shown in Fig. 1. On the encoder side, the spatial topology perception of GCNs is combined with the temporal dynamics modeling of GRUs, forming an innovative GCNGRU unit. The model synchronously captures the spatial dependencies among road network nodes and the temporal evolution of traffic states. The attention layer incorporates temporal decay correction to dynamically compute spatiotemporal correlations and adaptively adjust attention weights. The decoder is constructed using GRUs for multi-step prediction, and a novel input-feedback training strategy is devised to mitigate error accumulation by transferring historical attention vectors.

4.1 Encoder

We integrate GCNs and GRUs to construct the GCNGRU unit as the encoder. By replacing traditional matrix operations in the GRU with graph convolution operations, GCNGRU can simultaneously capture temporal dependencies and spatial correlations. Figure 2 provides a structural illustration of the model.

The computation process within the unit is formulated as follows:

$$z_t = \sigma(W_z[f(A, x_t), h_{t-1}] + b_z) \quad (2)$$

$$\mathbf{r}_t = \sigma(\mathbf{W}_r[f(\mathbf{A}, \mathbf{x}_t), \mathbf{h}_{t-1}] + \mathbf{b}_r) \quad (3)$$

$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{W}_h[f(\mathbf{A}, \mathbf{x}_t), (\mathbf{r}_t \odot \mathbf{h}_{t-1})] + \mathbf{b}_h) \quad (4)$$

$$\mathbf{h}_t = \mathbf{z}_t \odot \mathbf{h}_{t-1} + (1 - \mathbf{z}_t) \odot \tilde{\mathbf{h}}_t \quad (5)$$

where $f(\mathbf{A}, \mathbf{x}_t)$ denotes graph convolution operations, and $\mathbf{A}$ is the adjacency matrix. $\mathbf{z}_t$ and $\mathbf{r}_t$ represent the update gate and reset gate of the GCNGRU unit at time t , respectively. $\tilde{\mathbf{h}}_t$ is the candidate hidden state of the GCNGRU unit. $\mathbf{h}_{t-1}$ is the previous hidden state of the GCNGRU unit. σ is the activation function, and $\odot$ represents the Hadamard product.

Accordingly, taking the $(t - j)$ -th encoding time step as an example, the encoder's hidden state $\mathbf{h}_{t-j}$ is computed as:

$$\mathbf{h}_{t-j} = \begin{cases} \text{GCNGRU}_{enc}(\mathbf{X}_{t-j}, \mathbf{h}_0), & j = n - 1 \\ \text{GCNGRU}_{enc}(\mathbf{X}_{t-j}, \mathbf{h}_{t-j-1}), & j \in \{0, \dots, n - 2\} \end{cases} \quad (6)$$

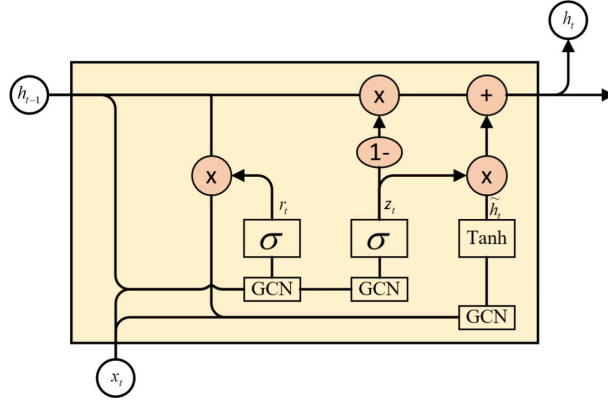


Fig. 2. GCNGRU unit structure.

4.2 Exponential-Decay Temporal Attention

To model evolving spatiotemporal features in traffic flow, we propose an exponential-decay temporal attention mechanism. The structural design of our exponential-decay temporal attention mechanism is depicted in Fig. 3.

Taking the $(t + i)$ -th decoding step as an example, the proposed mechanism is computed as follows:

First, we compute the similarity score e_{t-j}^{t+i} between the hidden vector $\mathbf{h}_{t+i}$ of the $(t + i)$ -th decoder step and the hidden vector $\mathbf{h}_{t-j}$ from the $(t - j)$ -th encoder time step, using the following scoring function:

$$e_{t-j}^{t+i} = \text{score}(\mathbf{h}_{t+i}, \mathbf{h}_{t-j}) = \mathbf{V}_e^T \tanh(\mathbf{W}_e[\mathbf{h}_{t+i}, \mathbf{h}_{t-j}]) \quad (7)$$

where $\mathbf{V}_e^T$ and $\mathbf{W}_e$ are learnable parameters. Then, the attention weight a_{t-j}^{t+i} for encoder step $t-j$ is formulated as:

$$a_{t-j}^{t+i} = \text{Softmax}\left(e_{t-j}^{t+i}\right) = \frac{\exp\left(e_{t-j}^{t+i}\right)}{\sum_{l=0}^{n-1} \exp\left(e_{t-l}^{t+i}\right)} \quad (8)$$

Next, we compute the decay correction factor $\varphi_{t+i}(k)$ using a time-exponential function:

$$\varphi_{t+i}(k) = e^{\left(-\frac{1}{n} \ln\left(\frac{1}{N_{\text{finish}}(t+i)}\right)k\right)}, \quad (t-n+1 \leq k \leq t) \quad (9)$$

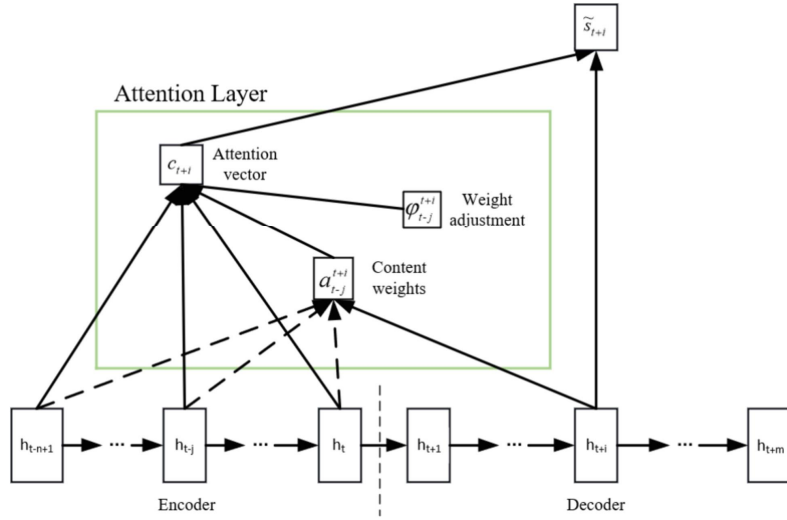


Fig. 3. Exponential-decay temporal attention mechanism

where n is the encoder sequence length, and $N_{\text{finish}}(t+i)$ the decay cutoff at step $t-j$. The variable k represents the decay time step $k = n - t + j$, which decreases as $t-j$ increases. For instance, at encoder step t (which corresponds to step n), $N_{\text{finish}}(t+i)$ is calculated as:

$$N_{\text{finish}}(t+i) = \lambda_{\min} + \frac{\lambda_{\max} - \lambda_{\min}}{n-1} (t+i-1) \quad (10)$$

where $\lambda_{\min}$ and $\lambda_{\max}$ are experimental parameters. We then adjust the attention weight a_{t-j}^{t+i} using the decay factor $\varphi_{t+i}(k)$, yielding the final weight $\tilde{a}_{t-j}^{t+i}$:

$$\tilde{a}_{t-j}^{t+i} = \varphi_{t+i}(k) \cdot a_{t-j}^{t+i} \quad (11)$$

Finally, the weighted sum of encoder hidden states produces the attention vector $\mathbf{c}_{t+i}$ for decoder time step $t+i$:

$$\mathbf{c}_{t+i} = \sum_{j=0}^{n-1} \tilde{a}_{t-j}^{t+i} \mathbf{h}_{t-j} \quad (12)$$

4.3 Decoder

We introduce a feedback-based training strategy that incorporates both the prior attention hidden state and the previous prediction as input to the present step. This mechanism enables the model to effectively retain historical context when computing attention weights, significantly improving long-term dependency modeling in sequential data processing.

Taking the decoding time step $t + i$ as an example, the computation of the decoder's hidden state vector $\mathbf{h}_{t+i}$ is given as follows:

$$\mathbf{h}_{t+i} = \begin{cases} GRU_{dnc}(\mathbf{X}_t; \mathbf{C}), & i = 1 \\ GRU_{dnc}(\mathbf{Input}_{dec}^{t+i}, \mathbf{h}_{t+i-1}), & i \in \{2, \dots, m\} \end{cases} \quad (13)$$

For $i \in \{2, \dots, m\}$, the decoder input $\mathbf{Input}_{dec}^{t+i}$ is computed as:

$$\mathbf{Input}_{dec}^{t+i} = \mathbf{W}_I [\hat{\mathbf{y}}_{t+i-1}; \tilde{\mathbf{s}}_{t+i-1}] \quad (14)$$

where $\mathbf{W}_I$ is a learnable parameter, $\hat{\mathbf{y}}_{t+i-1}$ represents the predicted output at decoding step $t + i - 1$, and $\tilde{\mathbf{s}}_{t+i-1}$ is the attention hidden state vector at step $t + i - 1$. The attention weight vector $\mathbf{c}_{t+i-1}$ and decoder hidden state $\mathbf{h}_{t+i-1}$ are concatenated and processed via a fully-connected layer with an activation function:

$$\tilde{\mathbf{s}}_{t+i-1} = \tanh(\mathbf{W}_{t+i-1} [\mathbf{c}_{t+i-1}; \mathbf{h}_{t+i-1}]) \quad (15)$$

where $\mathbf{W}_{t+i-1}$ is a learnable parameter.

5 Experiments

5.1 Experimental Setup

This study evaluates performance on two real-world traffic datasets. The SZ-taxi dataset records taxi trajectory data in Shenzhen from January 1 to January 31, 2015, with 156 main roads in the Luohu district. The Los-Loop dataset comprises traffic speed data collected by 207 loop detectors on Los Angeles highways between March 1 and March 7, 2012, sampled every 5 min. During training and testing, traffic flow values in each dataset are normalized to the $[0, 1]$ range using Min-Max normalization.

All models were trained on a server equipped with an NVIDIA Quadro M4000 GPU. We set the learning rate to 0.001 and batch size to 64, used MSE as the loss function, and adopted the Adam optimizer. The network's hidden layer contains 64 units. Data splitting follows an 80%-20% ratio for training-testing sets.

5.2 Baseline Models

Our approach is evaluated against seven existing baseline methods in the experiment.

- **LSTM** [9]: It is a specialized RNN designed to mitigate the vanishing gradient problem in recurrent neural networks.

- **GCN** [17]: A two-layer Graph Convolutional Network model.
- **STGCN** [3]: It captures spatiotemporal dependencies in traffic networks by integrating temporal convolutions and graph convolutions.
- **T-GCN** [4]: It combines GCNs with GRUs to model the spatial and temporal dependencies in traffic data sequentially.
- **Seq2Seq** [18]: A Seq2Seq model using GRU units for both encoder and decoder.
- **Seq2GCN** [19]: It uses GCGRU as an encoder and GRU as a decoder.
- **Seq2Seq-att** [20]: A GRU-based Seq2Seq model with an attention mechanism.

5.3 Evaluation Metrics

We adopt RMSE, MAE, and R^2 as evaluation metrics to assess the predictive performance of the RAGC-Seq2Seq model.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (16)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (17)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (18)$$

5.4 Analysis of Model Prediction Performance

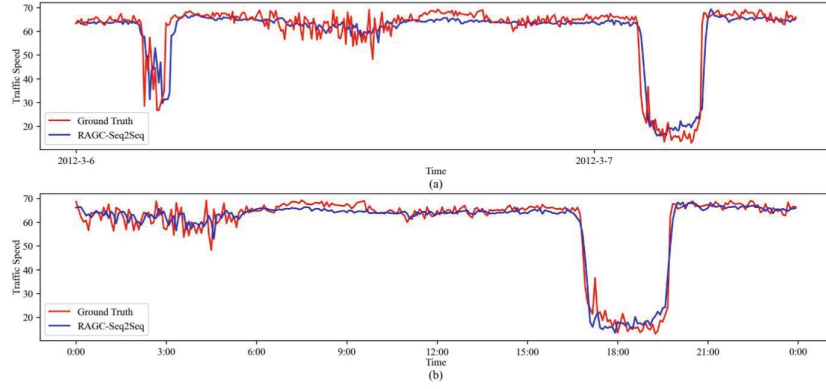
Table 1 compares the prediction accuracy between RAGC-Seq2Seq and baseline models across varying time horizons.

As shown in Table 1, models incorporating spatiotemporal dependencies in traffic data, such as Seq2GCN, STGCN, and T-GCN, generally outperform models that primarily focus on temporal features, such as LSTM, Seq2Seq, and Seq2Seq-Att. This observation emphasizes the necessity of incorporating spatiotemporal dependencies in traffic prediction tasks. Conversely, the GCN model, which solely models the spatial topology of the road network without accounting for temporal dependencies, demonstrates the lowest prediction accuracy across all baseline models.

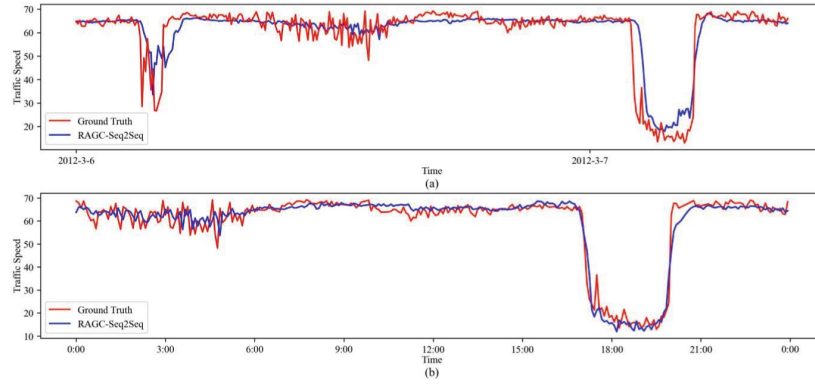
Overall, our approach is evaluated against seven existing baseline methods in the experiment, primarily due to its comprehensive integration of spatiotemporal features and historical attention information. It is also observed that prediction performance tends to decline as the prediction horizon increases, which aligns with practical experience that longer-term predictions are generally more challenging. The results further demonstrate that our model attains the highest evaluation metrics on both datasets for the 45-min and 60-min prediction horizons.

5.5 Visualization Analysis

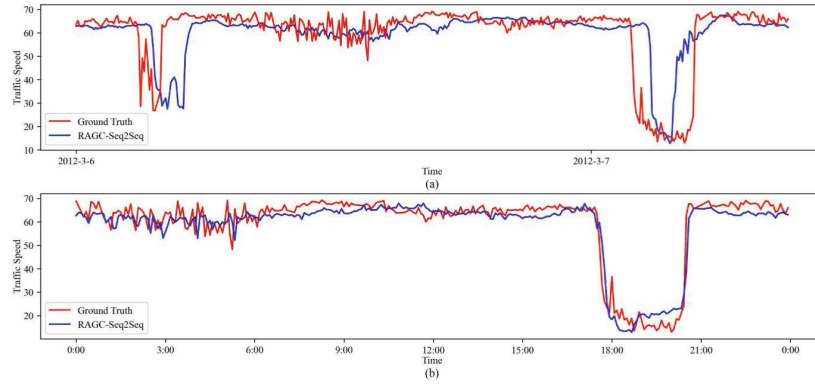
A representative road segment from the Los-Loop dataset was chosen to demonstrate our model's forecasting capability. The visual comparisons in Fig. 4 show predicted versus actual traffic patterns at 15-, 30-, and 60-min horizons, respectively.



(a) 15-minute prediction



(b) 30-minute prediction



(c) 60-minute prediction

Fig. 4. Visualization results of our model on the Los-Loop dataset.

Figure 4 shows that the divergence between predicted and observed traffic values widens as the prediction horizon lengthens, reflecting a performance degradation at longer prediction horizons. Despite traffic data's dynamic nature, the figures reveal that our model effectively captures the overall trend of traffic speed changes across both short-term (15 min) and long-term (60 min) horizons. The predicted values exhibit high overlap with actual values, and the prediction curves remain smooth and closely aligned with the real data, indicating strong predictive performance. However, when distinct peaks occur in actual speed data, the model's predictions tend to be less accurate in

Table 1. Performance comparison of different models on the SZ-Taxi and Los-Loop dataset.

Datasets	Methods	15 min			30 min			60 min		
		RMSE	MAE	R ²	RMSE	MAE	R ²	RMSE	MAE	R ²
SZ-Taxi	GCN	5.816	4.136	0.690	5.857	4.182	0.685	5.936	4.248	0.677
	LSTM	4.207	2.834	0.838	4.256	2.869	0.834	4.477	3.026	0.816
	STGCN	4.046	2.698	0.850	4.073	2.726	0.848	4.157	2.849	0.841
	T-GCN	4.083	2.743	0.847	4.102	2.760	0.846	4.172	2.790	0.840
	Seq2Seq	4.253	2.841	0.834	4.307	2.882	0.830	4.387	2.944	0.823
	Seq2SeqAtt	4.257	2.845	0.834	4.305	2.871	0.831	4.382	2.934	0.824
	Seq2GCN	4.105	2.750	0.845	4.122	2.768	0.844	4.167	2.803	0.841
	RAGC-Seq2Seq	4.054	2.731	0.849	4.080	2.735	0.847	4.107	2.765	0.845
Los-Loop	GCN	8.632	6.039	0.607	8.963	6.259	0.577	9.586	6.577	0.518
	LSTM	5.369	3.164	0.847	6.540	3.687	0.781	7.709	4.222	0.695
	STGCN	5.508	3.446	0.838	6.315	3.963	0.795	7.537	4.683	0.709
	T-GCN	5.238	3.241	0.858	6.195	3.786	0.802	7.436	4.563	0.717
	Seq2Seq	5.346	3.354	0.847	7.246	4.183	0.719	7.622	4.443	0.696
	Seq2SeqAtt	5.306	3.318	0.849	6.286	3.644	0.796	7.549	4.376	0.698
	Seq2GCN	5.285	3.303	0.850	6.211	3.524	0.801	7.542	4.362	0.701
	RAGC-Seq2Seq	5.069	3.204	0.865	6.049	3.623	0.808	7.101	4.128	0.736

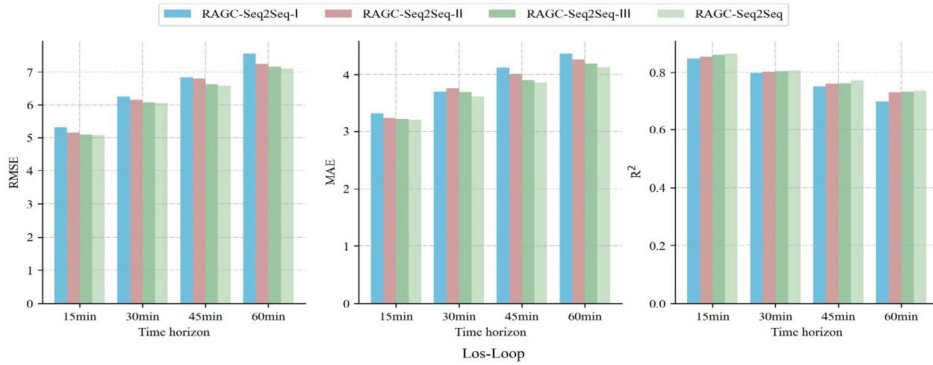


Fig. 5. The results of ablation experiments on the Los-Loop dataset

capturing these extremes. This reflects a common challenge faced by current traffic prediction models, largely due to the inherent randomness in traffic flow.

5.6 Ablation Experiment

Comprehensive ablation studies on the Los-Loop dataset were performed to systematically evaluate the architectural contributions of our proposed framework.

The ablated model variants are as follows:

- **RAGC-Seq2Seq-I**: This variant removes the GCN component from the encoder and uses only GRU as the encoder unit.
- **RAGC-Seq2Seq-II**: This variant eliminates the exponential decay factor in the temporal attention mechanism, retaining only the content-based similarity weight.
- **RAGC-Seq2Seq-III**: This variant removes the feedback input strategy in the decoder. It uses only the previous decoder output as input for the current decoding step, with all other structures unchanged.

Figure 5 presents the ablation study outcomes, where RAGC-Seq2Seq-I achieves the lowest performance across all three evaluation metrics on both datasets. This confirms the necessity of incorporating spatial dependencies in traffic data. RAGC-Seq2Seq-II exhibits a performance decline upon omitting the proposed temporal-exponential-decay attention mechanism. This indicates that the proposed mechanism improves the model's ability to capture temporal features in traffic data in conjunction with the conventional Luong attention mechanism, thereby improving predictive accuracy. Moreover, RAGC-Seq2Seq-III also shows reduced performance across all three evaluation metrics, suggesting that the feedback input training strategy proposed in this chapter is effective and contributes to improved prediction accuracy.

6 Conclusion

To address the limitations of conventional Seq2Seq models in extracting spatial features from traffic data and the information loss caused by independently computed attention vectors, we propose a multi-step traffic flow prediction model based on a reinforced attention graph convolutional network. In the encoder, the GCN-GRU unit is employed to encode the spatiotemporal characteristics of traffic data. The exponential-decay temporal attention mechanism is then applied to allow the decoder to focus on different positions in the encoder. In the decoder, the feedback-based training strategy concatenates the previous time step's attention hidden vector with its predicted output and feeds them into the current decoding step. Evaluation results on both SZ-taxi and Los-Loop datasets confirm the model's effectiveness in traffic prediction.

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